IMO 2026

Complete Solutions

GPT-5.6 Sol (default effort)

Independently graded: **28/42**

Autonomous run, no human intervention.
Solutions as written by the model; all six problems.

`github.com/deedy/imo-2026`

IMO 2026 — Problem 1

Solution by GPT-5.6 Sol

2026-07-18

Source: `results/gpt-5.6-sol/problem-01/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m, n)$ and $\frac{\operatorname{lcm}(m,n)}{\gcd(m,n)}$. He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1. (b) Prove that the value of M does not depend on the choices of Confucius.

Solution document

Status

solved

Problem

There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m, n)$ and $\frac{\operatorname{lcm}(m,n)}{\gcd(m,n)}$. He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1. (b) Prove that the value of M does not depend on the choices of Confucius.

Approaches tried

- **Lexicographic termination potential (successful):** track the total number of prime factors with multiplicity and, secondarily, the number of nonunit entries. Every move strictly decreases this pair lexicographically.
- **Prime-adic valuation invariant (successful):** for each prime p, a move changes the selected valuations (a, b) into $(\min(a, b), |a - b|)$, preserving the gcd of all p-adic valuations.

- **Product alone (insufficient):** the product of all entries is generally not invariant; it is divided by the selected pair's gcd. It helps motivate the first coordinate of the termination potential but cannot determine the final value.

Current best

Let the initial entries be $a_1, \dots, a_{2026}$. Every play terminates with exactly one nonunit, and its value is

$$\boxed{M = \prod_p p^{\gcd\left(v_p(a_1),\dots,v_p(a_{2026})\right)}},$$

where the product is over all primes and the gcd of an all-zero list is 0. This finite product is determined solely by the initial board.

Full proof

Let the entries currently on the board be $x_1, \dots, x_{2026}$. We first prove termination.

For a positive integer t, let $\Omega(t)$ denote the number of prime factors of t, counted with multiplicity, with $\Omega(1) = 0$. Define

$$S = \sum_{i=1}^{2026} \Omega(x_i), \qquad N = \#\{i : x_i > 1\}.$$

We compare pairs (S, N) lexicographically: the first coordinate is compared first, and the second coordinate is used only when the first coordinates are equal.

Suppose a move selects $m, n > 1$, and put $d = \gcd(m, n)$. The product of the two new entries is

$$d \cdot \frac{\operatorname{lcm}(m,n)}{d} = \operatorname{lcm}(m,n) = \frac{mn}{d}.$$

Since $\Omega(uv) = \Omega(u) + \Omega(v)$ for positive integers u, v, the contribution of these two positions to S changes from $$ (m)+ (n)

to [(m)+ (n)- (d). $$ If $d > 1$, then $\Omega(d) > 0$, so S strictly decreases. If $d = 1$, the new entries are 1 and mn. Thus S stays unchanged, while the two selected nonunits are replaced by exactly one nonunit, so N strictly decreases. Hence every move strictly decreases (S, N) lexicographically.

There cannot be infinitely many such decreases. Indeed, the nonnegative integer S can decrease only finitely many times, and during any interval in which S is fixed, the nonnegative integer N decreases at every move. Therefore every sequence of moves is finite.

When no further move is possible, at most one entry is greater than 1, since any two entries greater than 1 could be selected. On the other hand, a move can never leave no entry greater than 1: if the second new entry

$$\frac{\operatorname{lcm}(m,n)}{\gcd(m,n)}$$

equals 1, then $\operatorname{lcm}(m,n) = \gcd(m,n)$, which forces $m = n$, and the first new entry is then $m > 1$; if the second new entry is not 1, it is itself greater than 1. Since initially there are entries greater than 1, every configuration reached has at least one such entry. Consequently, every terminal configuration has exactly one entry $M > 1$. This proves part (a).

It remains to show that M is independent of the moves. Fix a prime p. For every positive integer x, write $v_p(x)$ for its exponent in the prime factorization of x. Define

$$G_p = \gcd(v_p(x_1), \dots, v_p(x_{2026})),$$

using the convention that the gcd of an all-zero list is 0.

We claim that G_p is invariant under every move. If

$$a = v_p(m), \qquad b = v_p(n),$$

then the valuations at the two selected positions become

$$v_p(\gcd(m, n)) = \min(a, b)$$

and

$$v_p\left(\frac{\operatorname{lcm}(m, n)}{\gcd(m, n)}\right) = \max(a, b) - \min(a, b) = |a - b|.$$

Assume without loss of generality that $a \le b$. Then

$$\gcd(\min(a, b), |a - b|) = \gcd(a, b - a) = \gcd(a, b).$$

Taking the gcd also with all valuations at the unchanged positions therefore gives the same value before and after the move. Thus G_p is indeed invariant.

Let the initial entries be $a_1, \dots, a_{2026}$. At the end, all entries except M equal 1, so their p-adic valuations are 0. The final value of the invariant is consequently

$$\gcd(v_p(M), 0, \dots, 0) = v_p(M).$$

Its initial value was $\gcd(v_p(a_1), \dots, v_p(a_{2026}))$. Hence, for every prime p,

$$v_p(M) = \gcd(v_p(a_1), \dots, v_p(a_{2026})).$$

By unique prime factorization,

$$M = \prod_p p^{\gcd(v_p(a_1), \dots, v_p(a_{2026}))}.$$

Only primes dividing at least one initial entry can have a nonzero exponent in this product, so it is finite. The formula involves only the initial entries and not the sequence of moves. Therefore the value of M is independent of all choices, proving part (b). $\square$

IMO 2026 — Problem 2

Solution by GPT-5.6 Sol

2026-07-18

Source: `results/gpt-5.6-sol/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC, respectively. Let points K and L be chosen inside triangles BMC and BNC, respectively, such that K lies inside the angle LBA, L lies inside the angle ACK, and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL. Prove that $OM = ON$.

Solution document

Status

solved

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC, respectively. Let points K and L be chosen inside triangles BMC and BNC, respectively, such that K lies inside the angle LBA, L lies inside the angle ACK, and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL. Prove that $OM = ON$.

Approaches tried

- Trigonometric/complex coordinates: successful. Coordinates of K, L follow from the sine rule in BMK, CNL; the two remaining angle conditions become reality relations.
- Equal powers: successful. A standalone complex-algebra lemma (`lemmas/complex-algebra.md`) shows that those reality relations imply that M, N have equal powers with respect to (AKL).
- Symbolic Laurent-polynomial checks (`code/symbolic.py`, `code/eliminate.py`, `code/certificate.py`) were used to check the algebra and orientations. An apparent orientation correction was rechecked: for counterclockwise ABC, the ray directions are $CL = \alpha + \pi + x$ and $CK = \alpha + \pi + x + z$.

Current best

The claim follows by normalizing $A = 0, B = 2, C = 2de^{i\alpha}$. Then $M = 1, N = de^{i\alpha}$, while the sine rule gives explicit complex coordinates for K, L. The angle order assumptions turn the remaining conditions into exactly the two hypotheses of the complex equal-power lemma, proving that M, N have equal powers with respect to (AKL). As both powers equal the squared distance to O minus the same circumradius squared, $OM = ON$.

Full proof

Reflect the whole configuration if necessary, and assume that A, B, C occur counterclockwise. We use complex coordinates, normalized by

$$A = 0, \qquad B = 2, \qquad C = 2n, \qquad n = de^{i\alpha},$$

where $d > 0$ and $\alpha = \angle BAC$. Thus

$$M = 1, \qquad N = n.$$

Set

$$x = \angle KBA = \angle ACL, \quad y = \angle LBK = \angle LNC, \quad z = \angle LCK = \angle BMK,$$

and abbreviate

$$X = e^{ix}, \qquad Y = e^{iy}, \qquad Z = e^{iz}, \qquad a = e^{i\alpha}.$$

We first find K. In triangle BMK we have

$$BM = 1, \qquad \angle MBK = x, \qquad \angle BMK = z.$$

Consequently $\angle BKM = \pi - x - z$, so the sine rule gives

$$MK = \frac{\sin x}{\sin(x+z)}.$$

The ray MK makes angle z with the positive real axis; hence

$$K = 1 + \frac{\sin x}{\sin(x+z)} Z. \qquad (1)$$

Likewise, in triangle CNL we have $CN = d$, $\angle NCL = x$, and $\angle CNL = y$. The ray NL has direction $\alpha - y$, and the sine rule gives

$$L = n\left(1 + \frac{\sin x}{\sin(x+y)} Y^{-1}\right). \qquad (2)$$

The denominators in (1)–(2) are nonzero because they are sines of angles supplementary to angles of the nondegenerate triangles BMK, CNL.

We next translate the ordering of the rays. Since K lies inside $\angle LBA$, when one turns inside that angle from BA toward BL, one meets BK first. Therefore BL has direction $\pi - x - y$. It follows that

$$(L-2)XY \in \mathbb{R}. \qquad (3)$$

At C, the ray CA has direction $\alpha+\pi$. Since $\angle ACL = x$, the ray CL has direction $\alpha+\pi+x$. Since L lies inside $\angle ACK$ and $\angle LCK = z$, the ray CK has direction $\alpha+\pi+x+z$. Thus the opposite vector $C-K = 2n-K$ has direction $\alpha+x+z$, and hence

$$\frac{2n-K}{aXZ} \in \mathbb{R}. \qquad (4)$$

We now apply the complex equal-power lemma proved in `lemmas/complex-algebra.md`; its short algebraic proof is reproduced here for self-containment. If p, q denote the complex coordinates of K, L, respectively, then, since

$$\frac{\sin x}{\sin(x+z)} = \frac{X-X^{-1}}{XZ-X^{-1}Z^{-1}}, \qquad \frac{\sin x}{\sin(x+y)} = \frac{X-X^{-1}}{XY-X^{-1}Y^{-1}},$$

formulas (1)–(2), together with (3)–(4), satisfy the hypotheses of that lemma. We recall its verification.

Write $t^* = \bar{t}$. The circle through $0, p, q$ has equation

$$w\bar{w} + uw + v\bar{w} = 0. \qquad (5)$$

Putting $w = p, q$ and applying Cramer's rule gives, with $\Delta = pq^* - qp^* \neq 0$,

$$u\Delta = -pp^*q^* + qq^*p^*, \qquad v\Delta = -pqq^* + qpp^*. \qquad (6)$$

The difference between the values of the left side of (5) at 1 and at n is, after multiplication by Δ,

$$\begin{aligned} H = (1-nn^*)\Delta + (1-n)(-pp^*q^* + qq^*p^*) \\ + (1-n^*)(-pqq^* + qpp^*). \qquad (7) \end{aligned}$$

The two reality relations (3)–(4) are equivalent to

$$(q-2)XY = (q^*-2)X^{-1}Y^{-1}, \qquad (8)$$
$$(2n-p)a^{-1}X^{-1}Z^{-1} = (2n^*-p^*)aXZ. \qquad (9)$$

Substitute

$$p = 1 + Z\frac{X-X^{-1}}{XZ-X^{-1}Z^{-1}}, \qquad q = da\left(1 + Y^{-1}\frac{X-X^{-1}}{XY-X^{-1}Y^{-1}}\right)$$

and their conjugates into (7). On using (8)–(9), and putting terms over the common denominator

$$(XZ-X^{-1}Z^{-1})^2(XY-X^{-1}Y^{-1})^2,$$

the numerator cancels in conjugate pairs: the four types are

$$(nq^*p^*, -n^*qp), \quad (npp^*q^*, -n^*pp^*q), \quad (qq^*p^*, -qq^*p), \quad (pq^*, -qp^*).$$

Thus $H = 0$. By (5)–(7), the powers of $1 = M$ and $n = N$ with respect to the circle through $0, p, q$, namely (AKL), are equal.

Let R be the radius of (AKL). The power of any point P with respect to this circle is $OP^2 - R^2$. Therefore

$$OM^2 - R^2 = ON^2 - R^2,$$

so $OM^2 = ON^2$. Both are nonnegative lengths, and hence

$$\boxed{OM = ON}.$$

IMO 2026 — Problem 3

Solution by GPT-5.6 Sol

2026-07-18

Source: `results/gpt-5.6-sol/problem-03/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1 and want to divide it between themselves. Liu Bang marks at most n points on the stick, and then Xiang Yu marks at most n points on the stick. The marked points are distinct. Then, the stick is cut at all marked points, creating a number of pieces. Afterwards, they take turns claiming any unclaimed piece of the stick, with Liu Bang going first. Each player's goal is to maximise the total length of their own pieces. For each n, determine the largest value c such that Liu Bang may guarantee a total length of at least c, regardless of Xiang Yu's play.

Solution document

Status

solved

Problem

Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1 and want to divide it between themselves. Liu Bang marks at most n points on the stick, and then Xiang Yu marks at most n points on the stick. The marked points are distinct. Then, the stick is cut at all marked points, creating a number of pieces. Afterwards, they take turns claiming any unclaimed piece of the stick, with Liu Bang going first. Each player's goal is to maximise the total length of their own pieces. For each n, determine the largest value c such that Liu Bang may guarantee a total length of at least c, regardless of Xiang Yu's play.

Approaches tried

- Reduced the claiming phase to the alternating sum of the final piece lengths after sorting them in nonincreasing order.
- Formulated the marking phase as a partition-refinement minimax problem.

- Used finite-grid brute force (`code/grid.py`) to identify and check the values $2/3, 4/7, 8/15, 16/31$ for small n; a SciPy optimization script was also written but unavailable dependencies prevented running it.
- Found the extremal initial partition: lengths proportional to $1, 2, 4, \dots, 2^n$.
- Proved the matching upper and lower bounds through the alternating-imbalance refinement lemma recorded in `lemmas/refinement.md`.
- Completed a rigor pass on the claiming phase using matching longest-piece strategies (rather than an unjustified dominance assertion), and included the column-sweep proof of the refinement lemma directly in the master proof.

Current best

The exact answer is

$$\boxed{c_n = \frac{2^n}{2^{n+1} - 1}}.$$

Liu attains it by making pieces proportional to $1, 2, 4, \dots, 2^n$. The refinement lemma both protects this geometric partition against Xiang's n cuts and lets Xiang reduce the imbalance of every other initial partition to the same universal bound.

Full proof

Put

$$D = 2^{n+1} - 1.$$

We shall prove that the answer is $2^n/D$.

1. The claiming phase

Let the final piece lengths in nonincreasing order be

$$x_1 \ge x_2 \ge \cdots \ge x_m.$$

The value of this claiming game is exactly

$$O = x_1 + x_3 + x_5 + \cdots. \qquad (1)$$

Indeed, if Liu always takes a longest remaining piece, then before his k-th turn at most $2k - 2$ pieces have disappeared, so the piece he takes has length at least x_{2k-1}. Thus he secures at least O. Conversely, if Xiang always takes a longest remaining piece, then before his k-th turn at most $2k - 1$ pieces have disappeared, so he takes a piece of length at least x_{2k}. He therefore secures at least $E = x_2 + x_4 + \cdots$, leaving Liu at most $1 - E = O$. These matching strategies prove (1), without requiring uniqueness among equal lengths.

Define the alternating imbalance

$$\Delta(x_1, \dots, x_m) = x_1 - x_2 + x_3 - x_4 + \cdots.$$

If $E = x_2 + x_4 + \cdots$, then $O + E = 1$ and $O - E = \Delta$, whence

$$O = \frac{1 + \Delta}{2}. \qquad (2)$$

2. The refinement lemma

We use the following lemma, proved in full in `lemmas/refinement.md`.

Alternating-imbalance refinement lemma. Let A be a finite multiset of positive numbers, and let $\Delta(A)$ denote the alternating sum after its members are arranged in nonincreasing order.

(i) If A has at most $q+1$ members and total sum T, it can be refined by at most q splits so that

$$\Delta \leq \frac{T}{2^{q+1}-1}. \qquad (3)$$

(ii) Every refinement of

$$\{d, 2d, 4d, \ldots, 2^q d\}$$

obtained by at most q splits has

$$\Delta \geq d. \qquad (4)$$

A split means replacing one member by two positive members having the same sum. Notice that marking a new point in an existing stick-piece is exactly one split. The lemma permits fewer than q splits, as required by the wording of the problem.

Proof of the lemma. Represent every member a by a vertical column of height a. If $N(t)$ columns reach level t, then

$$\Delta(A) = \int_0^\infty (N(t) \bmod 2)\, dt. \qquad (5)$$

Indeed, arranging the heights decreasingly shows that the integrand is 1 precisely on the intervals $(a_2, a_1), (a_4, a_3), \ldots$. A split replaces one column of height $u+v$ by columns of heights u, v.

We record the elementary sweep argument applied to these columns. With s cuts available and at most $s+1$ columns of total height T, sweep a horizontal line down through them, recording successive vertical portions alternately in two rows (according as the number of columns met is odd or even). Set $d = T/(2^{s+1}-1)$. When the first row has accumulated length d, cut the column crossed at that instant and put its lower portion in the other row; below that level the two rows are interchanged. Cancel equal portions occurring in both rows. What remains, apart from the initial portion of length d, consists of two arrays, each with at most s column-blocks and each of total length at most

$$\frac{T-d}{2} = (2^s - 1)d.$$

(The arrays are obtained by writing the successive swept portions as $u_1, u_3, \ldots$ and $u_2, u_4, \ldots$; cancellation pairs every common vertical portion, so their uncancelled totals add to $T - d$ and interchange at the cut.) Repeating the operation on the uncancelled array whose parity is odd proves by induction on s that its odd levels have total length at most d. The initial case $s = 0$ is one column. The first operation uses one cut and the induction uses at most $s-1$, proving (i) via (5).

For the reverse statement, sweep upward through columns of heights $d, 2d, \ldots, 2^s d$, divided mentally into strips of height d. A cut can interchange the two parity rows only above its level. Inductively, after the first $j+1$ columns have been swept, either one whole d-strip remains uncancelled in the odd row, or a cut has occurred in every one of these $j+1$ columns: adding the next column, whose 2^{j+1} strips consist of two identical copies of all preceding strips plus one parity interchange, preserves this alternative. At $j = s$, the second possibility requires at least $s+1$ cuts. With at most s cuts, an odd strip therefore remains, and (5) gives $\Delta \geq d$. This proves (ii). $\square$

3. Liu's guarantee

Liu uses all n marks to make $n+1$ pieces of lengths

$$\frac{1}{D}, \frac{2}{D}, \frac{4}{D}, \dots, \frac{2^n}{D}; \qquad (6)$$

their sum is 1. Xiang's marks perform at most n splits of this multiset. Lemma (ii), with $q = n$ and $d = 1/D$, gives

$$\Delta \geq \frac{1}{D}.$$

By (2), Liu's optimal-play total is at least

$$\frac{1}{2}\left(1 + \frac{1}{D}\right) = \frac{D+1}{2D} = \frac{2^n}{D}. \qquad (7)$$

4. Xiang's matching strategy

Whatever Liu does, his marks initially create a multiset A of at most $n+1$ positive lengths and total sum 1. By Lemma (i), Xiang can place at most n marks so that the resulting refinement satisfies

$$\Delta \leq \frac{1}{D}.$$

Equation (2) then shows that Liu can obtain at most

$$\frac{1}{2}\left(1 + \frac{1}{D}\right) = \frac{2^n}{D}. \qquad (8)$$

Thus (5) and (6) are matching lower and upper bounds, and consequently

$$\boxed{c_n = \frac{2^n}{2^{n+1} - 1}}.$$

IMO 2026 — Problem 4

Solution by GPT-5.6 Sol

2026-07-18

Source: `results/gpt-5.6-sol/problem-04/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ, then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Solution document

Status

solved

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ, then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Approaches tried

- **Integral-multiple descent (successful).** When $180^\circ = N\theta$, regard all angles in units of θ. First force both children to have an integral angle, then repeatedly split an integral angle

m into 1 and $m-1$.
- **Integer-entry interval lemma (successful).** For nonintegral $a+b+c=N$, after relabeling there is an integer k strictly between b and $a+b$; this specifies a legal cut whose two new opposite angles are $N-k$ and k.
- **Quotient-group invariant (successful).** Modulo $\theta\mathbb{Z}$, if the class of 180° is nonzero and all three current angle classes are nonzero, every cut has at least one child with the same property. Shan-Yu retains that child forever.
- **Computational sanity check.** Random normalized triangles for $2 \le N < 30$ verified the interval construction, positivity of all child angles, and the formulas for the two forced integral angles.

Current best

Mulan can guarantee victory exactly for

$$\boxed{\theta = \frac{180^\circ}{N} \quad \text{with an integer } N \ge 2.}$$

For these values she first forces an integral multiple of θ to survive regardless of Shan-Yu's choice, and then decreases that multiple until it is θ. For every other θ, Shan-Yu starts with an equilateral triangle and uses a nonzero-class invariant modulo $\theta\mathbb{Z}$ to avoid θ forever.

Full proof

We prove both directions.

1. Sufficiency

Suppose

$$180^\circ = N\theta$$

for an integer $N \ge 2$. Measure all angles in units of θ, so the angles of every triangle sum to N.

We first record an elementary fact.

Lemma 1. Let $a, b, c > 0$ satisfy $a+b+c = N$, where $N \ge 2$ is an integer. If none of a, b, c is an integer, then the variables can be relabeled so that, for some integer k,

$$b < k < a + b.$$

Proof. If one of the three numbers, say a, exceeds 1, take either of the other two numbers as b. Since b is not an integer,

$$b < \lceil b \rceil < b + 1 < b + a.$$

Thus $k = \lceil b \rceil$ works.

It remains to consider the case in which all three numbers are at most 1. As none is an integer, all are in fact less than 1, so $N = a + b + c < 3$. Since $N \ge 2$ is an integer, $N = 2$. Choose any one of the numbers as c and call the other two a, b. Then $b < 1$ and

$$a + b = 2 - c > 1.$$

Hence $k = 1$ works. $\square$

Now consider any current triangle. If it already has an angle θ, Mulan has won. If one of its normalized angles is an integer m, then necessarily $1 \le m \le N-1$. We will shortly explain how Mulan wins from such a position.

It therefore remains only to handle a triangle whose normalized angles a, b, c are all nonintegral. Relabel its vertices using Lemma 1, and choose an integer k such that

$$b < k < a + b.$$

Mulan cuts from the vertex having angle a to the opposite side, choosing the cut so that the part of angle a adjacent to the vertex having angle b is

$$x = k - b.$$

The inequalities above give $0 < x < a$, so the ray lies strictly inside the angle and meets the interior of the opposite side. Thus this is a legal cut.

In the child adjacent to the angle b, the angle at the point of the cut is

$$N - b - x = N - k,$$

whereas in the other child the angle at the point of the cut is

$$N - c - (a - x) = N - c - a + k - b = k,$$

because $a + b + c = N$. Therefore, no matter which child Shan-Yu retains, its triangle has an angle that is a positive integral multiple of θ. (Both k and $N - k$ are strictly between 0 and N, as they are angles of nondegenerate child triangles.)

Finally, suppose a current triangle has an angle $m\theta$, where m is an integer with $1 \le m \le N-1$. If $m = 1$, Mulan wins. If $m \ge 2$, she cuts from that vertex along the interior ray which splits the angle $m\theta$ into angles

$$\theta \quad \text{and} \quad (m-1)\theta.$$

Such a ray meets the interior of the opposite side, so the cut is legal. One child has an angle θ and the other has an angle $(m-1)\theta$. If Shan-Yu keeps the first, Mulan wins at the next stopping check; if he keeps the second, the positive integer m has decreased by 1. Repeating this argument forces victory after at most $m - 1$ further cuts. This proves sufficiency.

2. Necessity

Suppose now that

$$180^\circ \notin \theta\mathbb{Z}.$$

We show that Shan-Yu can prevent victory indefinitely.

Work in the additive quotient group

$$G = \mathbb{R}/(\theta\mathbb{Z}),$$

and denote the class of a real angle r by $[r]$. In particular, let

$$t = [180^\circ] \neq 0.$$

Call a triangle *safe* if none of its three angles has class 0 in G, equivalently, if none is an integral multiple of θ.

Shan-Yu can initially choose an equilateral triangle. It is safe: if $[60^\circ] = 0$, then $60^\circ = n\theta$ for some integer n, which would give $180^\circ = 3n\theta$, contrary to the assumption.

We prove that after every cut of a safe triangle, at least one child is safe. Let the classes of the parent angles be a, b, c. Then

$$a, b, c \neq 0, \qquad a + b + c = t \neq 0.$$

Suppose Mulan cuts from the vertex with angle class a, splitting that angle into classes u and v, where $u + v = a$. The two child triangles have angle-class triples

$$(u, b, c + v) \qquad \text{and} \qquad (v, c, b + u).$$

Indeed, for example, the third angle of the first child has class

$$t - b - u = (a + b + c) - b - u = c + v,$$

and similarly for the second child.

Assume for contradiction that neither child is safe. Since $b, c \neq 0$, the first child can contain a zero class only if

$$u = 0 \quad \text{or} \quad c + v = 0,$$

and the second can contain a zero class only if

$$v = 0 \quad \text{or} \quad b + u = 0.$$

There are four possible pairings. If $u = v = 0$, then $a = u + v = 0$. If $u = 0$ and $b + u = 0$, then $b = 0$. If $c + v = 0$ and $v = 0$, then $c = 0$. Each contradicts safety. In the sole remaining case,

$$c + v = 0 \quad \text{and} \quad b + u = 0,$$

so

$$t = a + b + c = (u + v) + b + c = (b + u) + (c + v) = 0,$$

again a contradiction. Hence at least one child is safe.

After every cut, Shan-Yu retains a safe child. Since an angle equal to θ has class 0, a safe triangle never triggers Mulan's victory. Thus Mulan cannot guarantee victory when $180^\circ \notin \theta\mathbb{Z}$.

Combining necessity and sufficiency, and using $0^\circ < \theta < 180^\circ$, the desired values are exactly

$$\boxed{\theta = \frac{180^\circ}{N} \quad (N = 2, 3, 4, \ldots).}$$

IMO 2026 — Problem 5

Solution by GPT-5.6 Sol

2026-07-18

Source: `results/gpt-5.6-sol/problem-05/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

solved

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Approaches tried

- **Iteration and rigidity (successful):** Substituting $x = f(y)$ forces an affine relation along every forward orbit. A symmetrized substitution then gives a quadratic modulus estimate for the displacement $g = f - \mathrm{id}$, from which monotonicity, continuity, and finally constancy follow.
- **Direct inequality reformulation (exploratory):** Rewriting both inequalities in terms of $g(x) - g(y)$ reveals error terms vanishing when $x = f(y)$, but the raw asymmetric bounds do not by themselves immediately compare all displacement classes. The iterate identity supplies the needed symmetry.
- **Sanity checking:** Random numerical tests verify both inequalities for translations $f(x) = x + c$ over many orders of magnitude; this is only a check, not part of the proof.

Current best

The complete set of solutions is

$$\boxed{f(x) = x + c \quad \text{for all } x > 0, \qquad c \geq 0.}$$

The proof first derives $f(f(t)) = 2f(t) - t$. For $g(t) = f(t) - t$, forward iteration gives $f^n(t) = t + ng(t)$ and hence $g \geq 0$. Comparing the original inequality at $(x, y) = (f(u), v)$ and $(f(v), u)$ gives a quadratic estimate on $|g(u) - g(v)|$. This forces f to be increasing, then g to be nondecreasing and continuous, and a partition argument forces g to be constant.

Full proof

Define

$$g(t) = f(t) - t \qquad (t > 0).$$

1. The iterate identity

In the given chain, set $x = f(y)$. Its two outer terms are both $f(y)$, since

$$\sqrt{\frac{f(y)^2 + f(y)^2}{2}} = f(y) \quad \text{and} \quad \sqrt{f(y)f(y)} = f(y).$$

Thus equality holds throughout, and

$$\frac{f(f(y)) + y}{2} = f(y).$$

Therefore

$$f(f(y)) = 2f(y) - y. \qquad (1)$$

In terms of g, equation (1) says

$$g(f(y)) = f(f(y)) - f(y) = f(y) - y = g(y). \qquad (2)$$

It follows by induction from (2) that, for every integer $n \geq 0$,

$$f^n(y) = y + ng(y). \qquad (3)$$

Every forward iterate belongs to $\mathbb{R}_{>0}$. If $g(y) < 0$, the right side of (3) is nonpositive for all sufficiently large n, a contradiction. Hence

$$g(y) \geq 0 \qquad \text{for all } y > 0. \qquad (4)$$

2. A quadratic comparison estimate

Fix $u, v > 0$. Apply the original chain with $x = f(u)$ and $y = v$. By (1), and by $v = f(v) - g(v)$, its middle numerator is

$$f(f(u)) + v = f(u) + g(u) + f(v) - g(v).$$

Writing $X = f(u)$, $Z = f(v)$, and $d = g(u) - g(v)$, we obtain

$$\sqrt{\frac{X^2 + Z^2}{2}} \geq \frac{X + Z + d}{2} \geq \sqrt{XZ}. \qquad (5)$$

The right-hand inequality in (5) gives

$$d \geq 2\sqrt{XZ} - X - Z = -(\sqrt{X} - \sqrt{Z})^2. \qquad (6)$$

Interchanging u and v in the same argument replaces d by $-d$, so (6) also gives

$$-d \geq -(\sqrt{X} - \sqrt{Z})^2.$$

Consequently

$$|g(u) - g(v)| \leq \left(\sqrt{f(u)} - \sqrt{f(v)}\right)^2. \qquad (7)$$

3. Monotonicity

We first prove that f is strictly increasing. Let $u > v$, and set

$$h = u - v > 0, \qquad d = g(u) - g(v), \qquad \Delta = f(u) - f(v) = h + d.$$

If $\Delta = 0$, (7) gives $d = 0$, contrary to $\Delta = h + d$ and $h > 0$.

Suppose instead that $\Delta < 0$. Then $d < -h$, while (7) yields

$$|d| \leq \left(\sqrt{f(u)} - \sqrt{f(v)}\right)^2 = |\Delta| \frac{|\sqrt{f(u)} - \sqrt{f(v)}|}{\sqrt{f(u)} + \sqrt{f(v)}} < |\Delta|.$$

On the other hand, since $\Delta = h + d < 0$,

$$|\Delta| = |d| - h < |d|,$$

a contradiction. Thus $\Delta > 0$, proving that f is strictly increasing.

Every iterate f^n is consequently strictly increasing. From (3), for $u > v$ and every positive integer n,

$$u + ng(u) = f^n(u) > f^n(v) = v + ng(v).$$

Hence

$$g(u) - g(v) > -\frac{u - v}{n}.$$

Letting $n \to \infty$ gives $g(u) \geq g(v)$. Therefore g is nondecreasing. (8)

4. Continuity

Fix $t > 0$. Since g is nondecreasing, its right limit

$$L = \lim_{s \downarrow t} g(s)$$

exists and is finite (for example, $g(t) \leq g(s) \leq g(t+1)$ when $t < s < t+1$). Put $J = L - g(t) \geq 0$ and $A = f(t) > 0$. As $s \downarrow t$, we have

$$f(s) = s + g(s) \longrightarrow t + L = A + J.$$

Passing to the limit in (7) gives

$$J \leq (\sqrt{A + J} - \sqrt{A})^2. \qquad (9)$$

If $J > 0$, then

$$(\sqrt{A+J} - \sqrt{A})^2 = J\frac{\sqrt{A+J} - \sqrt{A}}{\sqrt{A+J} + \sqrt{A}} < J,$$

contradicting (9). Hence $J = 0$, so g is right-continuous at t.

The left limit exists as well (when approaching within $\mathbb{R}_{>0}$). If

$$J = g(t) - \lim_{s \uparrow t} g(s) > 0,$$

then passing to the limit in (7) gives

$$J \le (\sqrt{f(t)} - \sqrt{f(t) - J})^2 < J,$$

again a contradiction. Thus g is also left-continuous. Therefore g, and hence $f(t) = t + g(t)$, is continuous on $\mathbb{R}_{>0}$. (10)

5. The displacement is constant

Fix $0 < a < b$. For a positive integer N, partition $[a, b]$ into N equal subintervals, with endpoints

$$x_i = a + \frac{i(b-a)}{N} \qquad (0 \le i \le N).$$

Because f is strictly increasing and g is nondecreasing, the quantities

$$\Delta_i = f(x_i) - f(x_{i-1}) > 0, \qquad d_i = g(x_i) - g(x_{i-1}) \ge 0$$

are nonnegative as indicated. By (7),

$$\begin{aligned} d_i &\le \left(\sqrt{f(x_i)} - \sqrt{f(x_{i-1})}\right)^2 \\ &= \frac{\Delta_i^2}{\left(\sqrt{f(x_i)} + \sqrt{f(x_{i-1})}\right)^2} \le \frac{\Delta_i^2}{4f(a)}. \end{aligned}$$

Summing and using telescoping gives

$$\begin{aligned} 0 \le g(b) - g(a) &\le \frac{1}{4f(a)} \sum_{i=1}^{N} \Delta_i^2 \\ &\le \frac{\max_i \Delta_i}{4f(a)} \sum_{i=1}^{N} \Delta_i \\ &= \frac{\max_i \Delta_i}{4f(a)} (f(b) - f(a)). \qquad (11) \end{aligned}$$

By (10), f is uniformly continuous on the compact interval $[a, b]$. The partition mesh tends to zero, so $\max_i \Delta_i \to 0$ as $N \to \infty$. Equation (11) therefore gives $g(b) = g(a)$. Since $a < b$ were arbitrary, g is constant on $\mathbb{R}_{>0}$.

Thus there is a constant c such that

$$f(x) = x + c.$$

By (4), $c \ge 0$.

6. Verification

Conversely, let $c \geq 0$ and define $f(x) = x + c$. Then f maps positive reals to positive reals, and

$$\frac{f(x) + y}{2} = \frac{x + (y + c)}{2} = \frac{x + f(y)}{2}.$$

For the two positive numbers x and $f(y)$, the root-mean-square/arithmetic-mean inequality and the arithmetic-mean/geometric-mean inequality give

$$\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{x + f(y)}{2} \geq \sqrt{xf(y)}.$$

Hence every such function works, completing the characterization.

IMO 2026 — Problem 6

Solution by GPT-5.6 Sol

2026-07-18

Source: `results/gpt-5.6-sol/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $a_1, a_2, a_3, \ldots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n, the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \ldots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n. (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y.)

Solution document

Status

solved

Problem

Let $a_1, a_2, a_3, \ldots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n, the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \ldots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n. (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y.)

Approaches tried

- Computed examples. Prime-power starts produce arithmetic progressions, while $a_1 = 15$ produces the periodic difference pattern of period 8 and total increment 30.
- Recast the sequence as the increasing enumeration of the integers non-coprime to every sequence term. This reduces the problem to proving that this set is periodic.
- Used prime-divisor sets and a finite-transversal lemma. The pairwise-intersecting family of prime-divisor sets has only finitely many necessary transversal patterns; consequently the admissible integers are a finite union of sets of multiples. This completes the proof.

Current best

The assertion is proved. There exist squarefree integers $b_1, \dots, b_s > 1$ such that an integer is non-coprime to every term precisely when it is divisible by at least one b_j. Taking $L = \operatorname{lcm}(b_1, \dots, b_s)$ and T equal to the number of admissible residue classes modulo L gives $a_{n+T} = a_n + L$ for every $n \geq 1$.

Full proof

For a positive integer $m > 1$, denote by $P(m)$ the finite nonempty set of its prime divisors. We first prove the set-theoretic lemma that will provide the needed finiteness.

Finite-transversal lemma. Let $\mathcal{F}$ be a nonempty, possibly infinite family of nonempty finite subsets of a set P. If every two members of $\mathcal{F}$ intersect, then there are finitely many nonempty finite sets $B_1, \dots, B_s \subseteq P$ such that, for every finite $X \subseteq P$,

$$X \cap F \neq \emptyset \quad \text{for every } F \in \mathcal{F} \quad \Longleftrightarrow \quad B_j \subseteq X \quad \text{for some } j.$$

Proof of the lemma. Call a finite set meeting every member of $\mathcal{F}$ a transversal. Every finite transversal contains an inclusion-minimal finite transversal, by successively deleting dispensable elements. We use the following elementary "finite blocker" fact.

> If a family of finite sets is pairwise intersecting and has a member C of size r, then it has only finitely many inclusion-minimal finite transversals.

For completeness, here is a proof. Construct a rooted search tree as follows. A node bears a finite set D. If D is a transversal, stop. Otherwise choose a member F_D disjoint from D, and attach the finitely many children $D \cup \{x\}$, $x \in F_D$. At the root first attach the r one-element sets $\{c\}$, $c \in C$; this loses no transversal, since every transversal meets the member C. Retain at a node only those branches which can still be extended to an inclusion-minimal finite transversal, and contract a node whenever an earlier chosen element has become dispensable.

This pruned tree is finite. Indeed, if it were infinite, König's lemma would give an infinite branch $x_1, x_2, \dots$. At stage k the set F_k selected there is disjoint from $\{x_1, \dots, x_k\}$. Since every F_k meets the fixed finite set C, one $c \in C$ belongs to infinitely many F_k. Insert c at its first such occurrence. Until the next occurrence nothing changes; at that next occurrence the element selected there is dispensable, because that selected set is already met by c. Repeating between consecutive occurrences shows that the branch has a contracted node, contrary to the pruning rule. Thus the tree is finite. Every minimal finite transversal follows a branch (at each nonterminal node it must contain some element of the selected set), and minimality prevents contraction; hence it is a terminal label. The finite blocker fact follows.

Apply this fact to any chosen $C \in \mathcal{F}$, which is a transversal because the family is pairwise intersecting. Let $B_1, \dots, B_s$ be all inclusion-minimal finite transversals. A finite X is a transversal exactly when it contains one of them. None is empty because $\mathcal{F}$ is nonempty. This proves the lemma. □

We apply the lemma to

$$\mathcal{F} = \{P(a_i) : i \geq 1\}.$$

If $i < j$, then the definition of a_j gives $\gcd(a_i, a_j) > 1$. Hence $P(a_i) \cap P(a_j) \neq \emptyset$, so $\mathcal{F}$ is pairwise intersecting. Obtain finite nonempty prime sets $B_1, \dots, B_s$ from the lemma, and put

$$b_j = \prod_{p \in B_j} p \qquad (1 \le j \le s).$$

For every positive integer $x > 1$, the lemma gives

$$\begin{aligned} \gcd(x, a_i) > 1 \text{ for every } i &\iff P(x) \cap P(a_i) \neq \emptyset \text{ for every } i \\ &\iff B_j \subseteq P(x) \text{ for some } j \\ &\iff b_j \mid x \text{ for some } j. \end{aligned}$$

Let

$$S = \{x \in \mathbb{Z}_{>0} : b_j \mid x \text{ for some } j\}.$$

Thus S is precisely the set of positive integers having gcd greater than 1 with every term a_i. In particular, every a_i belongs to S.

We next show that $a_1, a_2, \dots$ are exactly the members of S that are at least a_1, in increasing order. Certainly all terms belong to S. Conversely, suppose $x > a_1$ belongs to S. Because the sequence is strictly increasing, it is unbounded. Choose the unique n for which

$$a_n < x \le a_{n+1}.$$

Since $x \in S$, it has gcd greater than 1 with each of $a_1, \dots, a_n$. The minimality in the definition of a_{n+1} therefore gives $a_{n+1} \le x$. Hence $x = a_{n+1}$. Also $a_1 \in S$, proving the claim.

Finally, set

$$L = \operatorname{lcm}(b_1, \dots, b_s).$$

Membership in S depends only on the residue class modulo L, because each b_j divides L. Let T be the number of residue classes modulo L which belong to S. This is a positive integer: for example, the residue class 0 belongs to S.

For every integer y, each residue class modulo L occurs exactly once among

$$y + 1, y + 2, \dots, y + L.$$

Consequently exactly T of these integers lie in S. Moreover, $y \in S$ if and only if $y + L \in S$.

Take $y = a_n$. Then $a_n + L \in S$, and among the integers strictly greater than a_n and at most $a_n + L$ there are exactly T members of S. Since the terms of the sequence are the increasing enumeration of S from a_1 onward, the T-th term after a_n is therefore $a_n + L$. Thus

$$\boxed{a_{n+T} = a_n + L}$$

for every positive integer n, as required.